

EXPANSION OF FUNCTIONS

MACLAURIN'S THEOREM

If a function $f(x)$ is differentiable any number of times and can be expanded in a convergent series of terms of positive integral powers of x , then

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

PROOF: Let $f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots + a_n x^n + \dots$ where $a_0, a_1, a_2, a_3, \dots$ are to be determined. ... (1)

Differentiating (1) successively, we get

$$f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \dots + n a_n x^{n-1} + \dots$$

$$f''(x) = 2a_2 + 3 \cdot 2 a_3 x + 4 \cdot 3 a_4 x^2 + \dots + n(n-1) a_n x^{n-2} + \dots$$

$$f'''(x) = 3 \cdot 2 a_3 + 4 \cdot 3 \cdot 2 a_4 x + 5 \cdot 4 \cdot 3 a_5 x^2 + \dots + n(n-1)(n-2) a_n x^{n-3} + \dots$$

and, in general, $f^{(n)}(x) = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 \cdot a_n + \text{terms with positive powers of } x$.

At $x=0$, we have $f(0)=a_0$, $f'(0)=a_1$, $f''(0)=2 \cdot 1 a_2$, $f'''(0)=3 \cdot 2 \cdot 1 \cdot a_3$, ... in general, $f^{(n)}(0)=n! a_n$.

Substituting these values of $a_0, a_1, a_2, \dots$ in (1), we get

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!} f^{(n)}(0) \quad \dots (2)$$

which is the expansion of $f(x)$ in ascending powers of x and is known as Maclaurin's expansion of $f(x)$. The conditions under which the expansion (2) is valid, are

- (i) $f(x)$ and its successive derivatives must be finite and continuous in the given range of x
- (ii) the series on the right hand side of (2) must be convergent.

For the condition of convergence of the power series (2), if we write

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^{n-1}}{(n-1)!} f^{(n-1)}(0) + R_n \text{ where } R_n = \frac{x^n}{n!} f^{(n)}(\theta x) \quad 0 < \theta < 1.$$

then the remainder R_n should tend to 0 as n tends to ∞ .

Maclaurin's Expansion of Some Standard Functions

1. Expansion of $\sin x$. Let $f(x) = \sin x$, then $f^{(n)}(x) = \sin(x + n\pi/2)$ and $f^{(n)}(0) = \sin n\pi/2$

$$\therefore \sin x = \sum_1^{\infty} \left(\frac{x^n}{n!} \sin \left(\frac{n\pi}{2} \right) \right) \quad \text{or} \quad \boxed{\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots}$$

2. Expansion of $\cos x$

Since $\cos x = \frac{d}{dx} \sin x$, we differentiate the expansion of $\sin x$ term by term, hence

$$\boxed{\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots}$$

3. Expansion of e^x . If $f(x) = e^x$, $f^{(n)}(x) = e^x$ and $f^{(n)}(0) = 1$

$$\therefore \boxed{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots}$$

4. Expansion of $\log(1+x)$.

$$\text{If } f(x) = \log(1+x), \text{ then } f^{(n)}(x) = \frac{(-1)^{n-1} (n-1)!}{(1+x)^n} \quad \text{and} \quad f^{(n)}(0) = (-1)^{n-1} (n-1)!$$

$$\text{Thus, } \log(1+x) = \log 1 + \sum_1^{\infty} (-1)^{n-1} (n-1)! \frac{x^n}{n!} = \sum_1^{\infty} \frac{(-1)^{n-1} x^n}{n}$$

$$\text{or } \log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (|x| < 1) \Rightarrow \log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$

5. Expansion of $\tan x$.

[GGSIPU 2012]

If $f(x) = \tan x$, then $f'(x) = \sec^2 x$, $f''(x) = 2 \sec^2 x \tan x$, $f'''(x) = 2 \sec^4 x + 4 \sec^2 x \tan^2 x$,
hence $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$, $f'''(0) = 2$, $f^{(iv)}(0) = 0$, $f^{(v)}(0) = 16$, and so on.

$$\text{Therefore, } \boxed{\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots}$$

6. Binomial Expansion of $(1+x)^n$ for $|x| < 1$.

$$\boxed{(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots \quad (|x| < 1)}$$

TAYLOR'S THEOREM

Let $f(x)$ be a function of x and h be small. If the function $f(x+h)$ is capable of being expanded in a convergent series of terms of positive integral powers of h , then this expansion is given by

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots + \frac{h^n}{n!}f^{(n)}(x) + \dots$$

PROOF: Assume that $f(x+h) = A_0 + A_1h + A_2h^2 + A_3h^3 + \dots + A_nh^n + \dots$ where A 's are functions of $x \dots (1)$
Differentiating successively w.r.t. h , we get

$$f'(x+h) = A_1 + 2A_2 h + 3A_3 h^2 + 4A_4 h^3 + \dots + nA_n h^{n-1} + \dots$$

$$f''(x+h) = 2 \cdot A_2 + 3 \cdot 2 \cdot A_3 h + 4 \cdot 3 \cdot A_4 h^2 + \dots + n(n-1) A_n h^{n-2} + \dots$$

$$f'''(x+h) = 3 \cdot 2 \cdot A_3 + 4 \cdot 3 \cdot 2 \cdot A_4 h + \dots + n(n-1)(n-2) A_n h^{n-3} + \dots$$

and, in general, $f^{(n)}(x+h) = n(n-1)(n-2) \dots 3 \cdot 2 \cdot A_n + \text{terms in ascending powers of } h$.

Putting $h=0$, we get $f^{(n)}(x) = n! A_n$ so $A_n = \frac{f^{(n)}(x)}{n!}$, then (1) becomes

$$f(x+h) = f(x) + \frac{h}{1!} f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots + \frac{h^n}{n!} f^{(n)}(x) + \dots \quad \dots(2)$$

Its another form can be obtained by replacing x by a and h by $x-a$, so as to get

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots + \frac{(x-a)^n}{n!} f^{(n)}(a) + \dots \quad (3)$$

The conditions under which the above expansion is valid, are

- (i) the function $f(x)$ and its derivatives must be finite and continuous in the range of definition of $f(x)$.
- (ii) the series on the right hand side of (2) must be convergent for which the remainder term

$$R_n \rightarrow 0 \text{ as } n \rightarrow \infty \text{ where } R^n = \frac{h^n}{n!} f^{(n)}(x+\theta h) \text{ and } 0 < \theta < 1.$$

In the form (3) of Taylor's expansion, take $a=0$ then we have

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots$$

which is nothing but the **Maclaurin's expansion of $f(x)$** . Thus Maclaurin's expansion is a particular case of Taylor's expansion.

EXAMPLE 2.1. (a) Expand $\tan^{-1}x$ in powers of $(x-1)$.

[GGSIPU 2011]

(b) Expand $\log(1+\sin x)$ by Maclaurin's theorem.

[GGSIPU 2012]

SOLUTION: (a) The Taylor's expansion of $f(x)$ in powers of $(x-a)$, is

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots + \frac{(x-a)^n}{n!} f^{(n)}(a) + \dots$$

Here $f(x) = \tan^{-1}x$ and $a=1$ therefore, $f(1) = \pi/4$, and $f'(x) = \frac{1}{1+x^2}$ hence $f'(1) = \frac{1}{2}$.

Next $f''(x) = \frac{-2x}{(1+x^2)^2}$ hence $f''(1) = -\frac{1}{2}$,

and $f'''(x) = \frac{(1+x^2)^2(-2) - (-2x) \cdot 2(1+x^2) \cdot 2x}{(1+x^2)^4} = \frac{6x^2-2}{(1+x^2)^3}$, $\therefore f'''(1) = \frac{1}{2}$.

Therefore, we have $\tan^{-1}x = \frac{\pi}{4} + (x-1)\left(\frac{1}{2}\right) + \frac{(x-1)^2}{2!}\left(-\frac{1}{2}\right) + \frac{(x-1)^3}{3!}\left(\frac{1}{2}\right) + \dots$

or $\tan^{-1}x = \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{4}(x-1)^2 + \frac{1}{12}(x-1)^3 + \dots$ **Ans.**

(b) Let $f(x) = \log(1 + \sin x)$

$$f(0) = 0, \quad f'(x) = \frac{\cos x}{1 + \sin x} = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \quad \therefore f'(0) = 1,$$

$$f''(x) = -\frac{1}{2} \sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) \quad \therefore f''(0) = -\frac{1}{2} \sec^2 \frac{\pi}{4} = -1.$$

$$f'''(x) = \frac{1}{2} \sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right) \tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \quad \therefore f'''(0) = 1 \text{ and so on.}$$

$\therefore$ As Maclaurin's expansion we have

$$\log(1 + \sin x) = x + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(1) + \dots = x - \frac{x^2}{2} + \frac{x^3}{6} + \dots \quad \text{Ans.}$$

EXAMPLE 2.2.

(a) Show that $\log(1 + e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots$

[GGSIPU 2010]

(b) Find the Maclaurin's series expansion of $e^x \log(1 + x)$

[GGSIPU 2010]

SOLUTION: (a) Let $f(x) = \log(1 + e^x)$. By Maclaurin's theorem.

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \dots$$

Now $f(0) = \log 2, \quad f'(x) = \frac{e^x}{1 + e^x} \quad \therefore f'(0) = \frac{1}{2}, \quad f''(x) = \frac{e^x}{(1 + e^x)^2} \quad \therefore f''(0) = \frac{1}{4}$

$$f'''(x) = \frac{e^x(e^x - 1)}{(e^x + 1)^3} \quad \therefore f'''(0) = 0$$

and $f^{iv}(x) = \frac{e^x(1 + e^x)(2e^x - 1) - 3e^{2x}(e^x - 1)}{(e^x + 1)^4} \quad \therefore f^{iv}(0) = \frac{1}{8}, \dots$

Thus, $f(x) = \log(1 + e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \dots \quad \text{Hence proved.}$

(b) $e^x \log(1 + x) = \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} - \dots\right] \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots\right]$

$$= x + x^2\left(1 - \frac{1}{2}\right) + x^3\left(\frac{1}{3} + \frac{1}{2} - \frac{1}{2}\right) + x^4\left(-\frac{1}{4} + \frac{1}{3} - \frac{1}{4} + \frac{1}{6}\right) + x^5\left(\frac{1}{5} - \frac{1}{4} + \frac{1}{6} - \frac{1}{12} + \frac{1}{24}\right) + \dots$$

$$= x + \frac{x^2}{2} + \frac{x^3}{3} + x^4(0) + x^5\left(\frac{3}{40}\right) + \dots \quad \text{Ans.}$$

EXAMPLE 2.3.

(a) Expand the polynomial $2x^3 + 7x^2 + x - 1$ in powers of $(x - 2)$. [GGSIPU 2010; 2008]

(b) Expand $\tan x$ in powers of $\left(x - \frac{\pi}{4}\right)$ upto first four terms. [GGSIPU 2008]

SOLUTION: (a) Let $f(x) = 2x^3 + 7x^2 + x - 1$, then $f'(x) = 6x^2 + 14x + 1, f''(x) = 12x + 14, f'''(x) = 12$

Now, by Taylor's theorem for $f(x)$ about the point $x = 2$, we have

$$f(x) = f(2) + (x - 2)f'(2) + \frac{(x - 2)^2}{2!} f''(2) + \frac{(x - 2)^3}{3!} f'''(2) + \dots$$

Here $f(2) = 45, f'(2) = 53, f''(2) = 38, f'''(2) = 12$ hence

$$f(x) = 45 + 53(x - 2) + 19(x - 2)^2 + 2(x - 2)^3. \quad \text{Ans.}$$

(b) By Taylor's theorem the expansion of $f(x)$ about $x=a$, is

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots$$

Here $f(x) = \tan x$ and ' a ' = $\frac{\pi}{4}$ and $f\left(\frac{\pi}{4}\right) = \tan \frac{\pi}{4} = 1$.

$$f'(x) = \sec^2 x, \quad f''(x) = 2 \sec^2 x \tan x \quad \text{and} \quad f'''(x) = 2 \sec^4 x + 4 \tan^2 x \sec^2 x,$$

$$\therefore f'\left(\frac{\pi}{4}\right) = \sec^2 \frac{\pi}{4} = 2, \quad f''\left(\frac{\pi}{4}\right) = 2 \sec^2 \frac{\pi}{4} \tan \frac{\pi}{4} = 4, \quad \text{and} \quad f'''\left(\frac{\pi}{4}\right) = 2 \sec^4 \frac{\pi}{4} + 4 \tan^2 \frac{\pi}{4} \sec^2 \frac{\pi}{4} = 16$$

$$\therefore \tan x = 1 + \left(x - \frac{\pi}{4}\right) 2 + \frac{1}{2!} \left(x - \frac{\pi}{4}\right)^2 \cdot 4 + \frac{1}{3!} \left(x - \frac{\pi}{4}\right)^3 \cdot 16 \quad \text{upto first four terms}$$

$$= 1 + 2 \left(x - \frac{\pi}{4}\right) + 2 \left(x - \frac{\pi}{4}\right)^2 + \frac{8}{3} \left(x - \frac{\pi}{4}\right)^3. \quad \text{Ans.}$$

EXAMPLE 2.4.

(a) Prove that $\log(1+x+x^2+x^3+x^4) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} - \frac{4}{5}x^5 + \frac{x^6}{6} + \dots$

(b) Show that $(1+x)^x = 1 + x^2 - \frac{x^3}{2} + \frac{5}{6}x^4 - \frac{3}{4}x^5 + \frac{33}{40}x^6 + \dots$

SOLUTION: (a) Since $1+x+x^2+x^3+x^4 = \frac{1-x^5}{1-x}$ (as sum of a geometric series), we have

$$\log(1+x+x^2+x^3+x^4) = \log(1-x^5) - \log(1-x)$$

$$= \left[-x^5 - \frac{x^{10}}{2} - \frac{x^{15}}{3} - \frac{x^{20}}{4} - \dots \right] - \left[-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \frac{x^6}{6} - \dots \right]$$

$$= x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + x^5 \left(\frac{1}{5} - 1 \right) + \frac{x^6}{6} + \frac{x^7}{7} + \frac{x^8}{8} + \frac{x^9}{9} + x^{10} \left(\frac{1}{10} - \frac{1}{2} \right) + \dots$$

$$\text{or} \quad \log(1+x+x^2+x^3+x^4) = x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} - \frac{4x^5}{5} + \frac{x^6}{6} + \frac{x^7}{7} + \frac{x^8}{8} + \frac{x^9}{9} - \frac{2}{5}x^{10} + \dots$$

Hence Proved.

(b) We can write $(1+x)^x = e^{\log(1+x)^x} = e^{x \log(1+x)} = e^t$ where $t = x \log(1+x)$.

$$\text{Now,} \quad t = x \log(1+x) = x \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \right) = x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \frac{x^5}{4} + \dots$$

$$\therefore (1+x)^x = e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \frac{t^4}{4!} + \dots$$

$$= 1 + \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \frac{x^5}{4} + \frac{x^6}{5} - \dots \right) + \frac{1}{2!} \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \dots \right)^2 + \frac{1}{3!} \left(x^2 - \frac{x^3}{2} + \frac{x^4}{3} - \dots \right)^3 + \dots$$

$$= 1 + x^2 - \frac{x^3}{2} + x^4 \left(\frac{1}{3} + \frac{1}{2} \right) + x^5 \left(-\frac{1}{4} - \frac{1}{2} \right) + x^6 \left(\frac{1}{5} + \frac{1}{8} + \frac{1}{3} + \frac{1}{6} \right) + \dots$$

$$\text{or} \quad (1+x)^x = 1 + x^2 - \frac{x^3}{2} + \frac{5}{6}x^4 - \frac{3}{4}x^5 + \frac{33}{40}x^6 + \dots$$

Hence Proved.

EXAMPLE 2.5. Expand $\sin(m \sin^{-1} x)$ in ascending powers of x .

SOLUTION: Let $y = \sin(m \sin^{-1} x)$ then $y_1 = \cos(m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$... (1)

$$\begin{aligned} \text{or} \quad (1-x^2) y_1^2 &= m^2 \cos^2(m \sin^{-1} x) = m^2 (1-y^2) \\ \Rightarrow (1-x^2) 2y_1 y_2 - 2x y_1^2 &= -2m^2 y y_1 \quad \text{if } (y_1)_0 \neq 0 \\ \text{or} \quad (1-x^2) y_2 - x y_1 + m^2 y &= 0 \end{aligned} \quad \dots (2)$$

Now differentiating (2) on both sides, n times w.r.t. x , we get

$$\begin{aligned} (1-x^2) y_{n+2} + n(-2x) y_{n+1} - \frac{2n(n-1)}{2} y_n - [x y_{n+1} + n \cdot 1 \cdot y_n] + m^2 y_n &= 0 \\ \text{or} \quad (1-x^2) y_{n+2} - (2n+1)x y_{n+1} - (n^2-m^2) y_n &= 0 \end{aligned} \quad \dots (3)$$

At $x=0$ we have $(y_{n+2})_0 = (n^2-m^2) (y_n)_0$

Putting $x=0$ in (1) and (2) we get $y_1(0) = m$ and $(y_2)_0 = -m^2 y_1(0) = -m^3$.

$\therefore$ From (3), we get

$$\begin{aligned} (y_n)_0 &= [(n-2)^2 - m^2] (y_{n-2})_0 = [(n-2)^2 - m^2] [(n-4)^2 - m^2] (y_{n-4})_0 \\ &= [(n-2)^2 - m^2] [(n-4)^2 - m^2] [(n-6)^2 - m^2] (y_{n-6})_0 \\ &\dots \\ &= \begin{cases} [(n-2)^2 - m^2] [(n-4)^2 - m^2] \dots (2^2 - m^2) (y_2)_0 & \text{when } n \text{ is even} \\ [(n-2)^2 - m^2] [(n-4)^2 - m^2] \dots (1^2 - m^2) (y_1)_0 & \text{when } n \text{ is odd} \end{cases} \\ &= \begin{cases} [(n-2)^2 - m^2] [(n-4)^2 - m^2] \dots (4 - m^2) (-m^3) & \text{when } n \text{ is even} \\ [(n-2)^2 - m^2] [(n-4)^2 - m^2] \dots (1 - m^2) m & \text{when } n \text{ is odd} \end{cases} \end{aligned}$$

Thus, $(y_3)_0 = (1-m^2)m$, $(y_5)_0 = (3^2-m^2)(1-m^2)m$, ...

and $(y_4)_0 = (2^2-m^2)(-m^3)$, $(y_6)_0 = (4^2-m^2)(2^2-m^2)(-m^3)$, ...

$$\therefore y = 0 + mx + \frac{x^2}{2!} (-m^3) + \frac{x^3}{3!} m(1-m^2) + \frac{x^4}{4!} (4-m^2)(-m^3) + \frac{x^5}{5!} m(1-m^2)(3^2-m^2) + \dots \quad \text{Ans.}$$

EXAMPLE 2.6. Show that $\tan\left(\frac{\pi}{4} + x\right) = 1 + 2x + 2x^2 + \frac{8}{3}x^3 + \frac{10}{3}x^4 + \dots$ and hence calculate the value of $\tan 46^\circ$ correct to four decimal places, given that $\pi = 3.14159$.

SOLUTION: Let $f(x) = \tan\left(\frac{\pi}{4} + x\right)$, then $f(0) = 1$.

$$\therefore f'(x) = \sec^2\left(\frac{\pi}{4} + x\right) = 1 + [f(x)]^2 \quad \therefore f'(0) = 1 + 1 = 2$$

Next $f''(x) = 2f(x)f'(x)$, hence $f''(0) = 2f(0)f'(0) = 4$.

and $f'''(x) = 2f(x)f''(x) + 2\{f'(x)\}^2$ hence $f'''(0) = 2f(0)f''(0) + 2\{f'(0)\}^2 = 16$

Now, $f^{iv}(x) = 2f(x)f'''(x) + 2f'(x)f''(x) + 4f'(x)f''(x)$,

hence $f^{iv}(0) = 2f(0)f'''(0) + 6f'(0)f''(0) = 80$

$$\begin{aligned} \text{Therefore, } \tan\left(\frac{\pi}{4} + x\right) &= f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \dots \\ &= 1 + 2x + 2x^2 + \frac{8}{3}x^3 + \frac{10}{3}x^4 + \dots \end{aligned}$$

Hence the result.

Now taking $x = 1^\circ = \frac{\pi}{180} = \frac{3.14159}{180} = 0.01745$ we have $x^2 = 0.000345$ and $x^3 = 0.000005, \dots$

To achieve the desired accuracy we need consider only first four terms, hence

$$\tan 46^\circ = 1 + 2(0.01745) + 2(0.000345) + \frac{8}{3}(0.000005) = 1.0355$$

which is correct to four places of decimal.

EXAMPLE 2.7. Show that $\sin^{-1} x = x + \frac{1^2}{3!}x^3 + \frac{1^2 \cdot 3^2}{5!}x^5 + \frac{1^2 \cdot 3^2 \cdot 5^2}{7!}x^7 + \dots$
and hence find the approximate value of π .

SOLUTION: Let $y = \sin^{-1} x$ hence $y_1 = \frac{1}{\sqrt{1-x^2}}$ or $(1-x^2)y_1^2 = 1$.

Differentiating both sides w.r.t. x , gives

$$(1-x^2)2y_1y_2 - 2xy_1^2 = 0 \quad \text{or} \quad (1-x^2)y_2 - xy_1 = 0 \quad \text{as } y_1 \neq 0.$$

Differentiating the above relation n times using Leibnitz's theorem, we get

$$(1-x^2)y_{n+2} + {}^nC_1(-2x)y_{n+1} + {}^nC_2(-2)y_n - xy_{n+1} - {}^nC_1 1 \cdot y_n = 0$$

$$\text{or} \quad (1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$$

$$\text{At } x=0, \text{ we have } y_{n+2}(0) = n^2y_n(0) \quad \text{or} \quad y_n(0) = (n-2)^2y_{n-2}(0).$$

$$\text{and} \quad y_n(0) = (n-2)^2(n-4)^2y_{n-4}(0). \quad \text{Continuing the process, we get}$$

$$y_n(0) = (n-2)^2(n-4)^2(n-6)^2 \dots 2^2y_2(0), \quad \text{when } n \text{ is even}$$

$$= (n-2)^2(n-4)^2(n-6)^2 \dots 1^2y_1(0), \quad \text{when } n \text{ is odd.}$$

$$\text{therefore, } y_n(0) = 0 \quad \text{when } n \text{ is even (since } y_2(0) = 0)$$

$$= 1^2 \cdot 3^2 \cdot 5^2 \cdot \dots (n-2)^2 \quad \text{when } n \text{ is odd. (since } y_1(0) = 1)$$

Now, by Maclaurin's theorem, we have

$$\sin^{-1} x = 0 + x + \frac{1^2}{3!}x^3 + \frac{1^2 \cdot 3^2}{5!}x^5 + \frac{1^2 \cdot 3^2 \cdot 5^2}{7!}x^7 + \dots$$

$$= x + \frac{1}{2} \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \frac{x^5}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \frac{x^7}{7} + \dots$$

Here $-1 < x < 1$ and to obtain the value of π we put $x = 1/2$, to get

$$\frac{\pi}{6} = \frac{1}{2} + \frac{1}{6} \cdot \frac{1}{8} + \frac{3}{40} \cdot \frac{1}{32} + \frac{5}{112} \cdot \frac{1}{128} + \dots$$

$$\text{or} \quad \pi = 3 + \frac{1}{8} + \frac{9}{640} + \frac{30}{14336} = 3.141 \quad \text{approximately.} \quad \text{Ans.}$$

EXAMPLE 2.8. (a) Estimate the value of $\sqrt{10}$ correct to four places of decimal using Taylor's theorem. [GGSIPU 2003]

(b) Apply Taylor's theorem to estimate the value of $f(11/10)$ where

$$f(x) = x^3 + 3x^2 + 15x - 10.$$

[GGSIPU 2007; 2009]

SOLUTION: (a) Let $f(x) = \sqrt{x}$, then using Taylor's expansion we get

$$(x+h)^{1/2} = x^{1/2} + h \frac{d}{dx}(x^{1/2}) + \frac{h^2}{2!} \frac{d^2}{dx^2}(x^{1/2}) + \frac{h^3}{3!} \frac{d^3}{dx^3}(x^{1/2}) + \dots$$

Taking $x = 9$ and $h = 1$ in the above expansion, gives

$$\sqrt{10} = \left[3 + h \left(\frac{1}{2} x^{-1/2} \right) + \frac{h^2}{2!} \left(\frac{1}{2} \left(-\frac{1}{2} \right) x^{-3/2} \right) + \frac{h^3}{3!} \left(\frac{1}{2} \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) x^{-5/2} \right) + \dots \right]_{x=9, h=1}$$

$$= 3 + \frac{1}{2 \cdot 3} - \frac{1}{8 \cdot 27} + \frac{3}{3 \cdot 2 \cdot 8 \cdot 243} - \dots = 3.16227 \text{ approximately.} \quad \text{Ans.}$$

(b) By Taylor's expansion, we have

$$f(a+\delta x) = f(a) + \delta x f'(a) + \frac{(\delta x)^2}{2!} f''(a) + \frac{(\delta x)^3}{3!} f'''(a) + \dots \quad (1)$$

Here $f(x) = x^3 + 3x^2 + 15x - 10$ and $a=1$ and $\delta x=0.1$ $f(1) = 1^3 + 3 \cdot 1^2 + 15 \cdot 1 - 10 = 9$

$$f'(x) = 3x^2 + 6x + 15 \quad \therefore f'(1) = 3 + 6 + 15 = 24$$

$$f''(x) = 6x + 6, \quad \therefore f''(1) = 12$$

$$f'''(x) = 6 \quad \therefore f'''(1) = 6$$

$$\therefore f\left(\frac{11}{10}\right) = f\left(1 + \frac{1}{10}\right) = 9 + 0.1(24) + \frac{(0.1)^2}{2} 12 + \frac{(0.1)^3}{3!} (6) = 11.461. \quad \text{Ans.}$$

EXAMPLE 2.9. Apply Taylor's theorem to show that

$$\tan^{-1}(x+h) = \tan^{-1} x + h \sin \alpha \cdot \frac{\sin \alpha}{1} - (h \sin \alpha)^2 \cdot \frac{\sin 2\alpha}{2} + (h \sin \alpha)^3 \cdot \frac{\sin 3\alpha}{3} - \dots$$

[GGSIPU 2004]

where $\alpha = \cot^{-1} x$.

SOLUTION: Let $f(x) = \tan^{-1} x$, $\therefore f'(x) = \frac{1}{x^2 + 1} = \frac{1}{(x+i)(x-i)} = \frac{1}{2i} \left[\frac{1}{(x-i)} - \frac{1}{(x+i)} \right]$

Differentiating w.r.t. x , $(n-1)$ times, gives

$$f^{(n)}(x) = \frac{1}{2i} \left[\frac{(-1)^{n-1} (n-1)!}{(x-i)^n} - \frac{(-1)^{n-1} (n-1)!}{(x+i)^n} \right]. \text{ Since } \alpha = \cot^{-1} x \text{ or } x = \cot \alpha, \text{ we have}$$

$$f^{(n)}(x) = \frac{(-1)^{n-1} (n-1)!}{2i} \left[\frac{\sin^n \alpha}{(\cos \alpha - i \sin \alpha)^n} - \frac{\sin^n \alpha}{(\cos \alpha + i \sin \alpha)^n} \right]$$

$$= \frac{(-1)^{n-1} (n-1)!}{2i} \sin^n \alpha [(\cos n\alpha + i \sin n\alpha) - (\cos n\alpha - i \sin n\alpha)]$$

$$= (-1)^{n-1} \frac{2i}{(n-1)!} \sin^n \alpha \sin n\alpha \quad (\text{using de Moivre's Theorem})$$

By Taylor's Theorem $f(x+h) = f(x) + \frac{h}{1!} f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots + \frac{h^n}{n!} f^{(n)}(x) + \dots$

$$\therefore \tan^{-1}(x+h) = \tan^{-1} x + \frac{h}{1!} \sin \alpha \sin \alpha - \frac{h^2}{2!} \sin^2 \alpha \sin 2\alpha + \frac{h^3}{3!} \sin^3 \alpha \sin 3\alpha - \frac{h^4}{4!} \sin^4 \alpha \sin 4\alpha + \dots$$

$$= \tan^{-1} x + (h \sin \alpha) \frac{\sin \alpha}{1} - (h \sin \alpha)^2 \frac{\sin 2\alpha}{2} + (h \sin \alpha)^3 \frac{\sin 3\alpha}{3} - (h \sin \alpha)^4 \frac{\sin 4\alpha}{4} + \dots$$

Hence the result.

EXAMPLE 2.10. Expand $\cos^{-1} \left(\frac{x-x^{-1}}{x+x^{-1}} \right)$ in ascending powers of x . ($x > 0$)

SOLUTION: Let us put $x = \cot \theta$, then

$$\cos^{-1} \left(\frac{x-x^{-1}}{x+x^{-1}} \right) = \cos^{-1} \left(\frac{\cot \theta - \tan \theta}{\cot \theta + \tan \theta} \right) = \cos^{-1} \left(\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} \right)$$

$$= \cos^{-1}(\cos 2\theta) = 2\theta = 2 \cot^{-1} x = 2 \left(\frac{\pi}{2} - \tan^{-1} x \right) = \pi - 2 \tan^{-1} x.$$

We know that $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$ then general value of $\tan^{-1} x$ is written as

$$\tan^{-1} x = n\pi + x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

$$\begin{aligned} \text{therefore, } \cos^{-1} \left(\frac{x - x^{-1}}{x + x^{-1}} \right) &= \pi - 2 \left(n\pi + x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \right) \\ &= -(2n-1)\pi - 2x + \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{2}{7}x^7 - \dots \end{aligned} \quad \text{Ans.}$$